

Patient sex and outcomes in children with life-threatening hemorrhage

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Abstract

Background: Recent data suggest female sex imparts a survival benefit after trauma in adults. The independent associations between patient sex and age with outcomes have not been examined in children with life-threatening hemorrhage (LTH) from all etiologies.

Study Design and Methods: In a secondary analysis of a multicenter prospective observational study of children with LTH, Massive Transfusion in Children (MATIC), we analyzed if patient sex and age were associated with differences in severity of illness, therapies, and outcomes. Primary outcomes were 24 hour mortality and weight-adjusted transfusion volume during LTH. Kruskal–Wallis, chi-square testing, and multivariable linear regression were used for adjusted analyses.

Results: Of 449 children, 45% were females and 55% were males. Females were more commonly younger, white, and with less trauma as the etiology of LTH compared to males. Markers of clinical severity were similar between groups, except injury severity score (ISS) was higher in females in the trauma subgroup. In terms of resuscitative practices, females received greater weight-adjusted total transfusion volumes compared to males (76 (40–150) mL/kg vs. 53 (24–100) mL/kg), as well as increased red blood cells (RBCs), plasma, and platelets compared to males. After adjustment for confounders, female sex and age 0–11 years were independently associated with increased transfusion volume during LTH. There were no differences in mortality or adverse outcomes according to patient sex.

Conclusion: Patient sex and age may impact factors associated with LTH and therapies received. Studies in developmental hemostasis are needed to determine the optimal transfusion strategy for LTH according to patient sex and age.

KEYWORDS

life-threatening hemorrhage, outcomes, pediatric hemorrhage, sex differences

1 | INTRODUCTION

Life-threatening hemorrhage (LTH) is a significant contributor to mortality in children, with a fatality rate ranging from 20% to 51% depending on the underlying cause.^{1–4} Sex differences in hemostasis and physiologic response to shock may impact outcomes following LTH.^{5–9} Further understanding of sexual dimorphism in outcomes from LTH may elucidate differences in hemostatic, immune, or other biologic system functions that offer potential management targets for precision transfusion medicine and improvement of patient outcomes.¹⁰

The impact of sex on mortality in bleeding adults is variable in the literature. Several studies have reported a survival benefit for pubertal females compared to males and postmenopausal females after trauma, traumatic shock, or traumatic hemorrhage.^{8,11–14} Others have also reported decreased end-organ injury and sepsis in pubertal females with traumatic injury compared to males.^{12,15–19} However, other studies contradict these findings, reporting no difference in outcomes attributable to sex in trauma patients.^{20–22} A few studies of outcomes following adult cardiac surgery have reported mixed results, finding either no sex-based differences or finding women had increased short-term mortality.^{23,24} One study of adult patients with variceal bleeding found women had lower mortality after adjusting for age and clinical severity.²⁵

The data in children are far more limited; two pediatric-specific studies reported a survival benefit for pubertal females but not prepubertal females after trauma and traumatic shock.^{5,6} In this secondary analysis of the Massive Transfusion in Children (MATIC) study, a large multicenter prospective observational study of children with LTH of all etiologies, we hypothesized that there would be differences in resuscitative practices and outcomes attributable to patient sex after pediatric LTH.

2 | MATERIALS AND METHODS

2.1 | Study design and data

This is a post hoc determined secondary analysis of the MATIC study, which is comprised of data from 449 children aged 0–17 years with LTH from all causes at 24 tertiary care centers in the United States, Canada, and Italy from January 2014 to October 2018. LTH was defined as ≥ 40 mL/kg of total blood products over 6 h or if a patient was transfused after massive transfusion protocol (MTP) activation. Eligible patients were identified by the hospitals' blood bank records. For patients with multiple bleeding events, only the first event was analyzed. Patients

were categorized into traumatic, operative, or medical bleeding groups. Medical bleeding was defined as related to underlying medical condition (i.e., gastrointestinal bleeding). Operative bleeding was defined as related to a procedure and occurred during the procedure or immediately post-procedure. Traumatic bleeding was defined as related to blunt or penetrating trauma and included trauma patients requiring operative intervention for LTH. A full description of the MATIC study has been previously published.³

The primary outcomes were 24-hour mortality and weight-adjusted total transfusion volume during the LTH with secondary outcomes of 6-hour and 28-day mortality, acute kidney injury (AKI), acute respiratory distress syndrome (ARDS), and sepsis. Total transfusion volume was calculated by adding the weight-adjusted (in millilitre per kilogram) volume of red blood cells (RBCs), plasma, and platelets transfused during the LTH event. AKI was defined using KDIGO criteria,²⁶ and ARDS was defined according to the Berlin criteria.²⁷ Sepsis was defined as a suspected or proven infection caused by any pathogen (by positive culture, tissue stain, or polymerase chain reaction or a clinical syndrome associated with high probability of infection with evidence on clinical exam, imaging, or laboratory tests).

Additional data analyzed included demographics, pre-LTH vital signs, pre-LTH laboratory values, injury severity score (ISS) in patients with traumatic injury, Pediatric Risk of Mortality III score (PRISM III) score, Glasgow Coma Scale, and comorbid conditions and therapies (including central nervous system injury, cardiac arrest, cyanotic heart disease, preexisting renal disease, vasoactive drug administration, and use of dialysis, extracorporeal membrane oxygenation, or mechanical ventilation) prior to the LTH.

2.2 | Analysis

Statistical analyses were performed in RStudio (version 4.3.0; supplemental packages: “ggplot2,” “dplyr,” “haven,” “lme4,” “interactions,” “DHARMa,” “stats” [1–7]). Categorical data were described by percentage of total and compared using chi-square tests. Continuous variables were described using medians and interquartile ranges and compared using the Kruskal–Wallis test. Age was analyzed as a binary variable 0–11 years versus 12–17 years. Age 12 was chosen as the cut point to define puberty based on the most recent data of median age of menarche in the United States (12.3 years).²⁸ Given the heterogeneity of patients within the overall population, we explored subgroups based on bleeding etiology and age cohorts.

Multivariable logistic regression was used to analyze the association of sex and the log-transformed total volume of blood products transfused (in millilitre per kilogram). Univariate generalized linear regression was performed for each covariate with the log-transformed total volume of blood products transfused as the outcome in the overall cohort. Covariates related to volumes of platelets, plasma, RBCs, and cryoprecipitate transfused were excluded from the model because the total volume of blood was the summation of these variables. Unadjusted comparisons were performed for each etiology of LTH (trauma, operative, medical).

We used a backward stepwise selection approach with Akaike information criterion as the comparison criteria for feature selection.²⁹ For the overall population, we allowed all variables with a threshold of *p*-value less than 0.2 in the univariate model to be included in the full model. When variables were highly correlated with each other via Pearson correlation, only one was kept, chosen by minimizing missingness. We further excluded variables with missingness greater than 40% in the full model. Model diagnostics were performed using R package DHARMA.³⁰ McFadden's pseudo R² was calculated to check model performance.³¹

Within subgroup analysis, multivariable linear regression was performed for 28-day mortality within the 12–17 year trauma group. Univariate logistical regression for 28-day mortality was conducted within this subgroup and then multivariable model built as described above.

3 | RESULTS

Of the 449 children in the MATIC study, 45% were females and 55% were males. There were several baseline characteristic differences between sexes. Females were younger with a median (IQR) age of 6 (1–14) years compared to 9 (2–15) years in males (*p* = 0.003). The race composition of the sexes also differed (see Table 1) with the females having a higher percentage White and lower percentage Black children compared to males. There were also differences in the bleeding etiology with males having a majority trauma etiology. As such, LTH location also differed with males having a greater percentage take place in the emergency department. There were no differences in PRISM III scores, lowest hemoglobin prior to LTH, age-adjusted hypotension, or other clinical data collected (Table 1).

In subgroup analyses by bleeding etiology, females had higher ISS scores in the trauma subgroup: 33 (26–42) versus 26 (19–38) (*p* = 0.038) (Appendix A). All other baseline characteristics were similar between sexes in

each etiology category (data not shown for medical and operative subgroups).

During the LTH, females received higher volumes of platelets, plasma, and RBCs and as a result also received a higher total amount of blood products transfused (76 (40–150) mL/kg vs. 53 (24–100) mL/kg) (Table 1). There were no differences in crystalloid volume, or cryoprecipitate or antifibrinolytic administration between the sexes. In all sub-analyses of the bleeding etiologies, females received a numerically higher total transfusion volume; however, this was only statistically significant in the trauma subgroup. In the trauma subgroup, the total amount of blood products given during the LTH was 58 (32–108) versus 38 (20–66) mL/kg for females and males, respectively (*p* = 0.005); in the operative subgroup, it was 106 (53–207) versus 86 (37–138) mL/kg, respectively (*p* = 0.079), and in the medical subgroup, it was 66 (38–123) versus 47 (17–93) mL/kg, respectively (*p* = 0.052). Multivariable analyses for total transfusion volume for each etiology subgroup could not be completed due to too small of a sample size for the number of variables required for a well-adjusted analysis.

We conducted an adjusted analysis to assess whether potential confounders accounted for the total weight-adjusted transfusion volume differences between females and males. To normalize the data distribution to meet the underlying assumptions of the model, we used the log-transformed total weight-adjusted transfusion volume. In univariate analysis for the log-transformed total weight-adjusted transfusion volume, operative bleeding etiology was associated with higher volumes as well as many clinical indicators of medical complexity and severity (see Table 2). Black race and LTH location in the emergency department or intensive care unit (ICU) were associated with decreased transfusion volumes. Use of antifibrinolytics, cryoprecipitate, recombinant factor VII, and crystalloid volume (mL/kg) were correlated with higher transfusion volumes. After the final model adjusted for age, operative and traumatic bleeding etiologies, volume crystalloid (mL/kg), platelet:RBC ratio, lowest body temperature, pre-LTH partial thromboplastin time (PTT), antifibrinolytic use, and pre-LTH hemodialysis, female sex and age 0–11 years remained associated with higher transfusion volumes (Table 3). The increase in total weight-adjusted blood product transfusion volume between the male and female group was 31% with all other covariates held constant. The increase in total weight-adjusted blood product transfusion volume between children aged 12–17 years and 0–11 years was 48% with all other covariates held constant. Additionally, operative bleeding etiology, pre-LTH PTT, weight-adjusted crystalloid volume, antifibrinolytic use, and

TABLE 1 Comparison of males and females with LTH.

	<i>n</i>	Females (<i>n</i> = 202)	Males (<i>n</i> = 247)	<i>p</i> value
Demographics				
Age (years)	449	6 (1–14)	9 (2–15)	0.003
Race (<i>n</i>)	380			0.001
White		118 (71%)	124 (58%)	
Black		32 (19%)	77 (36%)	
Other		17 (10%)	12 (6%)	
Hispanic (<i>n</i>)	392	21 (12%)	22 (10%)	0.620
Clinical data				
Bleeding etiology: (<i>n</i>)	449			0.015
Trauma		78 (39%)	129 (52%)	
Operative		77 (38%)	76 (30%)	
Medical		47 (23%)	42 (17%)	
Event location: (<i>n</i>)	449			0.038
Emergency department		64 (32%)	96 (39%)	
Operating room		62 (31%)	63 (26%)	
CICU		18 (9%)	21 (8%)	
PICU		49 (24%)	56 (23%)	
NICU		8 (4%)	2 (1%)	
Other		1 (<1%)	9 (4%)	
Pediatric risk of mortality score	428	12 (6–21)	13 (6–23)	0.577
Lowest hemoglobin (g/dL)	423	9.5 (7.6–11.7)	9.8 (8–11.7)	0.441
Lowest platelets ($\times 10^9/L$)	333	146 (76–242)	163 (76–251)	0.546
Fibrinogen (mg/dL)	295	191 (131–271)	190 (141–248)	0.815
INR	250	1.5 (1.2–1.8)	1.5 (1.3–2.2)	0.733
PTT (seconds)	267	44 (32–78)	43 (31–100)	0.818
Lowest pH	317	7.2 (7.1–7.3)	7.2 (7.1–7.3)	0.937
Pre-life-threatening hemorrhage (LTH)	449	16 (8%)	20 (8%)	>0.999
Extracorporeal membrane oxygenation (<i>n</i>)				
Continuous vasoactive drug for hemodynamic support (<i>n</i>)	425	69 (35%)	69 (30%)	0.281
Pre-LTH mechanical ventilation (<i>n</i>)	432	111 (57%)	144 (61%)	0.410
Pre-LTH cardiac arrest (<i>n</i>)	438	31 (16%)	42 (18%)	0.668
Pre-LTH dialysis or CVVHD (<i>n</i>)	433	9 (5%)	11 (5%)	>0.999
Cyanotic heart disease (<i>n</i>)	424	26 (13%)	27 (12%)	0.713
Tachycardic for age (<i>n</i>)	422	122 (63%)	141 (62%)	0.806
Hypotension <3rd percentile for age (<i>n</i>)	443	117 (59%)	138 (57%)	0.790
Transfusions during LTH				
Duration of LTH (minutes)	448	227 (120–363)	186 (73–388)	0.066
Crystalloid volume (mL/kg)	449	16 (0–50)	18 (2–52)	0.502
Total blood product (mL/kg)	449	76 (40–150)	53 (24–100)	<0.001
RBC volume (mL/kg)	449	41 (20–76)	26 (13–49)	<0.001
Platelet volume (mL/kg)	449	10 (0–22)	5 (0–17)	0.005
Plasma volume (mL/kg)	449	19 (9–45)	14 (4–31)	0.006
Cryoprecipitate use (<i>n</i>)	449	77 (38%)	75 (30%)	0.104
Antifibrinolytic use (<i>n</i>)	449	28 (14%)	26 (11%)	0.350

(Continues)

TABLE 1 (Continued)

	<i>n</i>	Females (<i>n</i> = 202)	Males (<i>n</i> = 247)	<i>p</i> value
Outcomes				
Mortality at 6 hours (<i>n</i>)	445	35 (18%)	34 (14%)	0.337
Mortality at 24 hours (<i>n</i>)	445	49 (25%)	50 (20%)	0.332
Mortality at 28 days (<i>n</i>)	449	78 (39%)	90 (36%)	0.707
Acute kidney injury (<i>n</i>)	449	38 (19%)	45 (18%)	0.969
Acute respiratory distress syndrome (<i>n</i>)	449	43 (21%)	48 (19%)	0.713
Sepsis (<i>n</i>)	449	23 (11%)	21 (9%)	0.388

Abbreviations: CICU, Cardiac intensive care unit; CVVHD, Continuous veno-venous dialysis; LTH, Life-threatening hemorrhage; NICU, Neonatal intensive care unit; PICU, Pediatric intensive care unit.

platelet:RBC ratio remained independently associated with larger total weight-adjusted transfusion volume. Pre-LTH dialysis or continuous veno-venous hemodialysis was associated with decreased total weight-adjusted transfusion volume. Model diagnostics showed the model to meet all linear assumptions, and a McFadden's *R*-squared of 0.374 represented an excellent fit.

There were no differences according to patient sex in our outcomes of mortality, AKI, ARDS, and sepsis in the overall cohort (Table 1). In our analysis of the interaction of pubertal age and sex on outcomes, we found no differences in outcomes when stratified by age 0–11 years and 12–17 year cohorts. Adolescent females (aged 12–17 years) had numerically lower 28-day mortality in the trauma subgroup when compared to adolescent males (20% vs. 37%, $p = 0.152$), but differences remained insignificant with a coefficient (95% CI) of 0.663 (–0.591 to 1.917) ($p = 0.3$), when adjusted for ISS and PRISM scores.

4 | DISCUSSION

This analysis of children with LTH aimed to elucidate sex differences in mortality, adverse events, or resuscitative practices after LTH. In our analysis, no discernible sex differences in mortality, AKI, ARDS, and sepsis emerged at any time point in the overall cohort. On sub-analysis by age category, no sex differences in mortality emerged for 12–17-year-olds. Consequently, our findings do not directly support the notion of a protective effect of female sex regardless of age in the context of LTH in children; however, it is possible that there was a protective effect since mortality was similar between sexes despite females having more severe trauma and higher transfusion volumes of all blood components.

The literature evaluating sex-based outcomes in shock and hemorrhage is heavily skewed toward the trauma

population, with limited data regarding the role of patient sex in outcomes after nontraumatic causes of hemorrhage. Consequently, we conducted a sub-analysis specifically focusing on trauma patients within our study cohort. Females had more severe injury based on ISS, received larger total transfusion volumes during LTH, but had similar pre-LTH measures of shock or hemostatic dysfunction and no difference in mortality. Larger studies are required to allow for adjusted analyses to determine whether patient sex or age is independently associated with outcomes to include transfusion volume, morbidities, and mortality according to etiology of LTH in children.

Prior studies examining sex differences in trauma and traumatic shock have raised questions regarding whether observed protective effect of female sex in preclinical and clinical studies is attributable to variations in hemostatic differences or differences in other physiologic responses to stress such as immune system activation or unaccounted confounders. While the MATIC database does not contain functional coagulation markers, mechanistic hemostatic data, or hormone levels, sexual dimorphisms in coagulation are widely recognized to exist and age is an important modifier. One study of pediatric trauma patients demonstrated that sexual dimorphism in hemostasis as measured by thromboelastography emerges in pubertal children. In this study, females aged 12–17 years had a more hypercoagulable profile and less fibrinolysis compared to similarly aged males.⁷ Of note, there were no differences in mortality, ICU length of stay, and venous thrombosis incidence between sexes in either age cohort. In vitro studies have demonstrated sex hormones affect platelet aggregation.^{32,33} Another in vitro study demonstrated greater improvement of clotting time and angle on thromboelastography from addition of platelets and cryoprecipitate from premenopausal female donors compared to males.³⁴ A recent study in a pig traumatic brain injury-hemorrhage model found that animals

TABLE 2 Unadjusted analysis of factors associated with total transfusion volume (in millimeter per kilogram) during life-threatening hemorrhage.

	Univariate association with transfusion volume		
	<i>N</i>	Coefficient (95% CI)	<i>p</i> -value
Demographics			
Age, years	449		
0–11		-	
12–17		−0.63 (−0.87 to −0.39)	<0.001
Sex (<i>n</i>)	449		
Female		-	
Male		−0.54 (−0.77 to −0.30)	<0.001
Race (<i>n</i>)	380		
White		-	
Black		−0.36 (−0.64 to −0.07)	0.016
Other		0.32 (−0.17 to 0.81)	0.2
Hispanic (<i>n</i>)	392	−0.07 (−0.48 to 0.34)	0.7
Clinical data			
Bleeding etiology: (<i>n</i>)	449		
Medical		-	
Trauma		−0.12 (−0.43 to 0.19)	0.5
Operative		0.65 (0.32 to 0.97)	<0.001
Event location: (<i>n</i>)	449		
CICU		-	
Emergency department		−1.1 (−1.5 to −0.65)	<0.001
Operating room		−0.31 (−0.75 to 0.14)	0.2
PICU		−0.58 (−1.0 to −0.13)	0.013
NICU		−0.17 (−1.0 to 0.69)	0.7
Other		−1.4 (−2.4 to −0.30)	0.013
Pediatric risk of mortality score	428	0.01 (0 to 0.02)	0.045
Lowest hemoglobin (g/dL)	423	−0.04 (−0.08 to −0.01)	0.024
Lowest body temperature (°C)	339	−0.09 (−0.16 to −0.02)	0.01
PTT (seconds)	267	0 (0 to 0.01)	0.028
INR	250	0 (−0.07 to 0.07)	>0.9
Pre-life-threatening hemorrhage (LTH) extracorporeal membrane oxygenation (<i>n</i>)	449	0.56 (0.12 to 1.0)	0.013
Continuous vasoactive drug for hemodynamic support (<i>n</i>)	425	0.52 (0.26 to 0.79)	<0.001
Pre-LTH mechanical ventilation (<i>n</i>)	432	0.3 (0.05 to 0.54)	0.017
Pre-LTH cardiac arrest (<i>n</i>)	438	−0.12 (−0.44 to 0.2)	0.2
Pre-LTH dialysis or CVVHD (<i>n</i>)	433	−0.44 (−1.0 to 0.13)	0.13
Cyanotic heart disease (<i>n</i>)	424	1.1 (0.74 to 1.4)	<0.001
Tachycardic for age (<i>n</i>)	422	0.38 (0.13 to 0.63)	0.003
Hypotension <3rd percentile for age (<i>n</i>)	443	0.03 (−0.22 to 0.27)	0.8

(Continues)

TABLE 2 (Continued)

	Univariate association with transfusion volume		
	N	Coefficient (95% CI)	p-value
Transfusion data (during LTH)			
Duration of LTH (minutes)	448	0 (0 to 0)	0.5
Crystalloid volume (mL/kg)	449	0 (0 to 0.01)	<0.001
RBC volume (mL/kg)	449	1.7 (1.7 to 1.7)	<0.001
Platelet volume (mL/kg)	449	0.03 (0.02 to 0.03)	<0.001
Plasma volume (mL/kg)	449	2.7 (2.6 to 2.8)	<0.001
Cryoprecipitate use (n)	449	1.2 (1.0 to 1.5)	<0.001
Antifibrinolytic use (n)	449	0.67 (0.3 to 1.0)	<0.001
Recombinant factor VII use (n)	447	0.85 (0.49 to 1.2)	<0.001

Abbreviations: CVVHD, Continuous veno-venous dialysis; LTH, Life-threatening hemorrhage; PTT, Partial thromboplastin time.

TABLE 3 Adjusted analysis of factors associated with total transfusion volume (in millimeter per kilogram) during life-threatening hemorrhage.

	Multivariable association with Total transfusion volume (mL/kg)		
	N	Coefficient* (95% CI)	p-value
Age, years	449		
0–11		-	
12–17		-0.475 (-0.728 to -0.223)	<.001
Sex, male (n)		-0.309 (-0.541 to -0.077)	.009
Bleeding etiology: (n)	449		
Medical		-	
Trauma		-0.043 (-0.376 to 0.290)	.801
Operative		0.376 (0.065 to 0.687)	.018
Lowest body temperature (°C)	339	-0.049 (-0.111 to 0.013)	.12
PTT (seconds)	267	0.003 (0.001 to 0.006)	.002
Pre-LTH dialysis or CVVHD (n)	433	-0.558 (-1.002 to -0.113)	.014
Crystalloid volume (mL/kg)	449	0.005 (0.003 to 0.007)	<.001
Antifibrinolytic use (n)	449	0.455 (0.089 to 0.821)	.015
Platelet:RBC ratio	413	0.522 (0.184 to 0.859)	.002

Note: *Coefficients from the multivariable model can be interpreted as the percent change in log weight-adjusted transfusion volume relative to the reference category (binary or categorical covariate) or for a 1 unit increase in the continuous covariates.

Abbreviations: CVVHD, Continuous veno-venous hemodialysis; LTH, Life-threatening hemorrhage; PTT, Partial thromboplastin time.

randomized to treatment with an estrogen ester had better survival than animals that received placebo.³⁵ These data suggest hemostatic development is modified by patient sex, potentially partially regulated by sex hormones, and may confer a protective benefit to pubertal females facing LTH. From this information, we had hypothesized that females aged 12–17 years in our study would have improved survival compared to males in both age categories and females aged 0–11 years; however, our

findings did not support it. Within the overall cohort and within bleeding etiology subgroups, there were no differences in mortality between males and females of either age group.

While our data did not find a mortality difference between adolescent males and females, the 12–17-year-old age category was associated with a 48% reduction in total transfusion volume compared to 0–11-year-old children, which may reflect influences of sex hormones or

other differences in developmental hemostasis. This difference remained significant in the trauma subgroup, but not in the medical or operative bleeding etiology groups. Association of age of child and transfusion volumes is not widely reported, and there are conflicting data. In an unadjusted analysis of children with injuries at a US Army Combat Hospital, children aged ≤ 8 years had increased unit-based volume of RBCs compared to children and adults > 8 years, although also had worse ISS and greater mortality.³⁶ However, a national study of pediatric trauma patients found that older children (5–11 years) and adolescents (12–14 years) were more likely to receive a massive transfusion (> 40 mL/kg in first 24 h of admission) compared to young children (< 5 years).² Beyond physiologic differences in hemostasis to explain our results, one explanation of our results of greater weight-based transfusion volumes in young children could be weight-based dosing applied in young children compared to unit-dosing of blood products employed in adolescents. Further studies in developmental hemostasis may illuminate differences in resuscitation needs based on age.

Interestingly, the few studies in adults that have evaluated the association between patient sex and outcomes after severe hemorrhage have found comparable results to our study with no differences in mortality attributable to patient sex. One study of trauma patients who had a massive transfusion protocol activation found no difference in mortality; however, this study may have been limited by the small number of female patients, comprising only 14% of the study population.³⁷ Similarly, a secondary analysis of the Pragmatic Randomized Optimal Platelet and Plasma Ratios (PROPPR) trial found no difference in mortality among male and female trauma patients predicted to require massive transfusion.³⁸ Two other studies of adult trauma patients who met study-defined criteria for clinically relevant bleeding found males had more events of infection and multiorgan failure but no mortality difference attributable to sex.^{15,39} It is possible that the presence of hemorrhage and the shock that occurs with it negates any protective effect of female sex after injury.

Beyond differences in survival attributable to biological differences between sexes, we were interested in evaluating for differences in treatment based on patient sex. Equity in medical treatment has been a goal prompting recent investigation in sex, gender and other socioeconomic disparities. Recent work has revealed there are gaps in treatment by patient sex. A sub-analysis of the Clinical Randomization of an Antifibrinolytic in Significant Hemorrhage (CRASH-2 and CRASH-3) trials found that although women had similar risk reduction of

mortality from tranexamic acid, they were less likely to receive it.⁴⁰

In our study, females received a larger total weight-adjusted transfusion volume, and this was attributable to females receiving larger weight-adjusted transfusion volumes of RBCs, plasma, and platelets. After adjusting for presentation severity, female sex remained an independent predictor of increased transfusion volumes. Bias of providers conducting resuscitation could alter transfusion practices if females were perceived to be at greater risk. Other explanations could lie in clinical severity or patient response to treatment not captured by this study; for example, a less robust response of vital signs in female patients to initial transfusions could prompt continued transfusions and would not be represented in our data as real-time heart rate and blood pressure measurements were not captured. It is also notable that the duration of LTH was numerically higher with a median of 40 minutes longer in females compared to males, although duration was not found to be associated with increased total transfusion volume on univariate analysis. Since platelets and plasma are predominantly donated by males due to TRALI risk mitigation strategies that limit females with a history of pregnancy from donating,⁴¹ it is possible that the blood used from male donors is not as effective in female recipients as in males. Increased mortality according to donor–recipient sex differences has been reported with RBC transfusions,^{42,43} although there are other studies contradicting this relationship.⁴⁴ Additional pediatric-specific studies should assess the impact of blood product characteristics on survival and organ failure outcomes.

It is also noteworthy that antifibrinolytic use and higher platelet:RBC ratios were independently associated with higher weight-based transfusion volumes in multivariable analysis. These practices would be expected to reduce need for blood transfusions by improving hemostasis. Antifibrinolytic use has been shown in multiple studies in children to reduce intra-operative blood transfusions.^{45–47} Patients with adequate clinical response to initial transfusion measures would be less likely to receive additional products, and so this association may be result of residual confounding despite our analysis controlling for markers of patient acuity.

Limitations to our study include that it is observational in nature as well as a secondary analysis of a dataset; thus, analysis was limited to existing data. Given heterogeneous sample of patients and missing data for some variables, we were underpowered for some comparisons. Small sample numbers precluded adjusted analyses of total transfusion volume of each bleeding etiology subgroup. We predetermined age 12 as the cut point to

determine adolescence but acknowledge there are limited data for age of developmental changes in hemostasis and this may differ between sexes. Furthermore, many factors influence transfusion practices and outcomes in patients with LTH and it is difficult to capture all vital signs and other clinical data imperative in making resuscitation decisions. While we attempted to account for confounding variables in our adjusted analyses, there could be other data that explain these results. In particular, this study did not incorporate functional coagulation assays such as thromboelastography given rarity of use by the included pediatric centers at time of data collection; data regarding functional coagulation and fibrinolysis could add valuable information to future studies of sex differences in hemostasis. This study does not account for the lack of consistency of massive transfusion protocols across centers, and specific policies are not known for sites included. It is most likely that the transfusion methods were empiric ratios of products rather than point-of-care testing-directed given limited use of viscoelastic testing.

5 | CONCLUSION

Patient sex and age 0–11 years were associated with requiring larger total weight-based transfusion volumes during LTH; however, no sex-based differences in mortality or adverse events were seen. Studies in developmental hemostasis are needed to determine the optimal transfusion strategy for LTH according to patient sex and age.

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CONFLICT OF INTEREST STATEMENT

Phillip C. Spinella is a consultant for Cerus and Hema-next, on the advisory board for Octapharma and Haima, and is a co-founder and CMO for Kalocyte. Cassandra Josephson discloses financial support as a consultant at Westat, Medtronic, and Immucor. Remaining authors have no COI disclosures.

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APPENDIX A: COMPARISON OF MALES AND FEMALES WITH TRAUMATIC LTH

	<i>n</i>	Females (<i>n</i> = 78)	Males (<i>n</i> = 129)	<i>p</i> value
Demographics				
Age (years)	207	9 (5–15)	12 (4–16)	0.276
Race (<i>n</i>)	173			0.099
White		40 (68%)	58 (51%)	
Black		18 (30%)	54 (47%)	
Other		1 (2%)	2 (2%)	
Hispanic (<i>n</i>)	181	7 (11%)	9 (8%)	0.644
Clinical data				
Event location: (<i>n</i>)	207			0.647
Emergency department		55 (70%)	84 (65%)	
Operating room		12 (15%)	23 (18%)	
PICU		11 (14%)	20 (16%)	
Other		0	2 (2%)	
Injury mechanism (<i>n</i>)	207			0.121
Blunt		54 (69%)	71 (55%)	
Penetrating		19 (24%)	51 (40%)	
Blunt and penetrating		5 (6%)	7 (5%)	
Injury severity score	187	33 (26–42)	26 (19–38)	0.015
Glasgow coma score	173	6 (3–14)	3 (3–12)	0.267
Pediatric risk of mortality score	193	10 (5–18)	12 (5–24)	0.478
Lowest hemoglobin (g/dL)	189	9.6 (8.3–11.9)	10.1 (8.5–12.1)	0.448
Lowest platelets ($\times 10^9/L$)	138	209 (148–284)	216 (150–282)	0.869
Fibrinogen (mg/dL)	122	188 (139–232)	182 (143–228)	0.783
INR	98	1.4 (1.3–1.7)	1.5 (1.3–2.1)	0.35
PTT (seconds)	107	38 (30–61)	37 (28–94)	0.867
Lowest pH	104	7.1 (7.0–7.2)	7.2 (7.0–7.2)	0.749
Continuous vasoactive drug for hemodynamic support (<i>n</i>)	191	9 (12%)	17 (14%)	0.804
Pre-LTH mechanical ventilation (<i>n</i>)	196	37 (49%)	69 (57%)	0.367
Pre-LTH cardiac arrest (<i>n</i>)	201	15 (20%)	23 (18%)	1
Tachycardic for age (<i>n</i>)	187	41 (59%)	68 (58%)	1
Hypotension <3rd percentile for age (<i>n</i>)	203	41 (54%)	69 (54%)	1
Transfusions during LTH				
Duration of LTH (minutes)	207	202 (90–363)	175 (75–319)	0.18
Crystalloid volume (mL/kg)	207	27 (3–60)	20 (2–58)	0.739
Total blood product (mL/kg)	207	58 (32–108)	38 (20–66)	0.005
RBC volume (mL/kg)	207	36 (18–66)	21 (12–36)	0.003
Platelet volume (mL/kg)	207	7 (0–15)	3 (0–11)	0.054
Plasma volume (mL/kg)	207	16 (4–34)	11 (4–23)	0.074
Cryoprecipitate use (<i>n</i>)	207	22 (28%)	27 (21%)	0.306
Antifibrinolytic use (<i>n</i>)	207	11 (14%)	11 (9%)	0.304

(Continues)

	<i>n</i>	Females (<i>n</i> = 78)	Males (<i>n</i> = 129)	<i>p</i> value
Outcomes				
Mortality at 6 h (<i>n</i>)	205	14 (18%)	21 (16%)	0.84
Mortality at 24 h (<i>n</i>)	205	20 (26%)	30 (23%)	0.746
Mortality at 28 days (<i>n</i>)	207	26 (33%)	48 (37%)	0.679
Acute kidney injury (<i>n</i>)	207	3 (4%)	9 (7%)	0.541
Acute respiratory distress syndrome (<i>n</i>)	207	13 (17%)	18 (14%)	0.742
Sepsis (<i>n</i>)	207	9 (12%)	6 (5%)	0.115

Abbreviations: LTH, Life-threatening hemorrhage; PTT, Partial thromboplastin time.